QML based traffic management in 6G networks

1st Anish Kundu
Department of CSE
KIIT University
Bhubaneswar, India
anishkundu28092002@gmail.com

2nd Anishesh Mitra
Department of CSE
KIIT University
Bhubaneswar, India
anishesh2004@gmail.com

3rd Subhajit Chakraborty
Department of CSE
KIIT University
Bhubaneswar, India
subhajitchakrabortyguddu@gmail.com

***Abstract*—Sixth-generation (6G) networks demand intelligent, ultra-low-latency systems for real-time network slice classification, yet traditional machine learning faces scalability limitations at the 6G edge. This paper proposes a Hybrid Variational Quantum Circuit (VQC) Neural Network to classify traffic into three ITU-defined slices—eMBB, URLLC, and mMTC—trained on the Kim & Choi 5G Traffic Dataset (IEEE DataPort, 2023), comprising 328 hours of real Wireshark captures across 15 applications with eight statistical features extracted per one-second observation window. The architecture combines a classical pre-processing layer, a 6-qubit VQC with four strongly entangling layers, and a classical post-processing classifier operating in a 64-dimensional quantum feature space ($2^6 = 64$). Benchmarked against Random Forest (92%), SVM (81.78%), and Gradient Boosting (92%) baselines, the hybrid VQC achieves 72.89% classification accuracy after five epochs with URLLC precision of 93.6%, demonstrating that QML's exponential representational capacity and native noise tolerance position it as a viable candidate for intelligent 6G network slice management on future quantum hardware.**

***Index Terms*—6G networks, network slicing, quantum machine learning, variational quantum circuits, traffic classification, hybrid quantum-classical architecture, eMBB, URLLC, mMTC, network slice management**

I. Introduction

The transition from fifth-generation (5G) to sixth-generation (6G) wireless networks is expected to redefine global mobile communication infrastructure, targeting peak throughputs exceeding 1 Tbps, end-to-end latency below 100 microseconds, and ubiquitous connectivity for hundreds of billions of heterogeneous devices [1]. Central to realizing this vision is network slicing—a software-defined networking technique that partitions a shared physical infrastructure into multiple isolated logical networks, each tailored to a specific traffic class. The International Telecommunication Union (ITU) has standardized three primary slice categories: enhanced Mobile Broadband (eMBB) for high-throughput services, Ultra-Reliable Low-Latency Communications (URLLC) for mission-critical applications, and massive Machine-Type Communications (mMTC) for dense Internet of Things (IoT) deployments [2].

Accurate real-time classification of network traffic into these slices is fundamental to autonomous 6G resource management. Classical machine learning methods, including Support Vector Machines (SVM), Random Forests, and Gradient Boosting, have demonstrated strong performance on traffic classification benchmarks [3]. However, as traffic volumes scale and feature spaces grow with network heterogeneity, these approaches face increasing computational overhead, particularly under the strict latency constraints imposed by URLLC slices at the 6G edge [4].

Quantum Machine Learning (QML) offers a theoretically motivated alternative. Variational Quantum Circuits (VQCs) encode input data into quantum state spaces of dimension 2^n, where n denotes the number of qubits, enabling exponential representational capacity relative to qubit count [5]. This property, combined with the inherent probabilistic structure of quantum systems, positions VQCs as strong candidates for high-dimensional pattern recognition in telecommunications. Nevertheless, purely quantum models remain constrained by the coherence limitations and gate error rates of Noisy Intermediate-Scale Quantum (NISQ) devices [6]. Hybrid quantum-classical architectures, which integrate classical preprocessing with a variational quantum core, provide a viable path toward leveraging quantum advantages within present hardware constraints [7].

This paper presents a Hybrid VQC Neural Network for real-time 6G network slice classification. The model integrates a classical pre-processing layer, a 6-qubit VQC with four strongly entangling layers trained via the parameter-shift rule through a custom PyTorch autograd bridge, and a classical post-processing classifier. Training and evaluation are conducted on the Kim & Choi 5G Traffic Dataset (IEEE DataPort, 2023), comprising 328 hours of real Wireshark packet captures from a live South Korean 5G network across 15 applications, with eight statistical flow-level features extracted per one-second observation window. The model is benchmarked against Random Forest, SVM (RBF kernel), and Gradient Boosting baselines, achieving 72.89% classification accuracy after five epochs with URLLC precision of 93.6%, establishing a proof-of-concept for quantum-assisted traffic management in future 6G networks.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and feature engineering methodology. Section IV details the hybrid model architecture and training procedure. Section V presents experimental results and comparative analysis. Section VI discusses findings and limitations. Section VII concludes the paper.

II. Related Work

A. Classical Machine Learning for Traffic Classification

Statistical machine learning methods have been widely applied to network traffic classification. Random Forests and Gradient Boosting ensembles have demonstrated strong performance on flow-level features, achieving 92% and 93% accuracy respectively on the Kim & Choi 5G Traffic Dataset [1]. SVMs with RBF kernels have also been effective, reaching 81.67% on the same benchmark [2]. However, these methods scale poorly under the strict latency constraints of URLLC slices in 6G networks [3].

B. Quantum Machine Learning in Communications

Variational Quantum Circuits (VQCs) encode classical data into quantum states of dimension 2^n, offering exponential representational capacity relative to the number of qubits . Hybrid quantum–classical architectures combine classical pre-processing with a variational quantum core, enabling operation within NISQ device constraints while preserving potential quantum advantages. However, existing QML studies in telecommunications remain limited to synthetic datasets and have not addressed real-world 5G slice classification under the ITU-defined three-class taxonomy of eMBB, URLLC, and mMTC. This work directly addresses this gap.

III. DATASET AND FEATURE ENGINEERING

A. Dataset

This work uses the Kim & Choi 5G Traffic Dataset (Kaggle, 2022), comprising real Wireshark CSV captures segmented into one-second windows across 15 applications [7]. A balanced subset of 1200 samples (400 per class) is drawn from 2250 eMBB, 5550 URLLC, and 1587 mMTC windows. Slice mapping follows ITU definitions as shown in Table I.

TABLE I
Dataset Applications and Slice Mapping

Category Folder	Slice (Label)	Applications
Stored_Streaming	eMBB (0)	Netflix, Amazon Prime, YouTube
Live_Streaming	eMBB (0)	AfreecaTV, Naver NOW, YouTube Live
Video_Conferencing	eMBB (0)	Zoom, MS Teams, Google Meet
Game_Streaming	URLLC (1)	GeForce Now, KT GameBox
Online_Game	URLLC (1)	Battleground, Teamfight Tactics
Metaverse	mMTC (2)	Zepeto, Roblox

B. Feature Engineering

Eight statistical flow-level features are extracted per window: throughput (kbps), average packet size (bytes), inter-arrival time mean and standard deviation (μs), flow packet count, downlink/uplink ratio, and bitrate mean and standard deviation (kbps). Features are standardized to zero mean and unit variance, then angle-encoded into 4 qubits via $\theta_i = \arctan(x_i)$ prior to quantum circuit ingestion. The dataset is split 75/25 into 900 training and 300 test samples as shown in Table II.

TABLE II
Extracted Flow-Level Features

Feature Name	Description
throughput_kbps	Total bytes × 8 / 1000 in window
avg_pkt_size_bytes	Mean packet length (bytes)
inter_arrival_mean_us	Mean inter-packet gap (μs)
inter_arrival_std_us	Std dev of IAT — jitter proxy
flow_pkt_count	Packets per second in window
dl_ul_ratio	Downlink bytes / uplink bytes
bitrate_mean_kbps	5-window rolling mean of throughput
bitrate_std_kbps	5-window rolling std of throughput

C. Class Balancing

The raw window counts across classes are significantly unequal: eMBB produces 2,250 windows, URLLC produces 5,550 windows, and mMTC produces only 1,587 windows. To prevent the model from being biased towards the majority class, we under-sample each class to the size of the smallest class, resulting in 300 windows per class and 900 total samples. The final dataset is split 75/25 into training (675 samples) and test (225 samples) sets using stratified sampling to preserve class proportions.

IV. Proposed Architecture

A. Overall Hybrid Framework

The proposed model adopts a hybrid quantum–classical architecture comprising three sequential modules: a classical pre-processing block, a variational quantum circuit (VQC) layer, and a classical post-processing block. The input features are first standardized using zero-mean, unit-variance normalization and subsequently clipped to the range $[-\pi, \pi]$ to satisfy the angular encoding requirements of the quantum layer.

B. Classical Pre-processing Block

The pre-processing block takes eight standardized features as input and applies a fully connected linear layer ($8 \rightarrow 6$ neurons), followed by a Tanh activation function and a batch normalization layer. The Tanh activation constrains the outputs to the range $(-1, 1)$, which is suitable for angle encoding into qubit rotation gates. The output of this block is a six-dimensional vector, with one value per qubit.

C. Variational Quantum Circuit

The VQC operates on six qubits, resulting in a $2^6 = 64$-dimensional Hilbert space. The circuit consists of three stages. First, the AngleEmbedding layer encodes the six pre-processed features as R_y rotation angles on each qubit, transforming the initial $|0\rangle$ state into a data-dependent quantum state. Second, $N_{\text{LAYERS}} = 4$ applications of StronglyEntanglingLayers perform parameterized R_z–R_y–R_z rotations on each qubit,

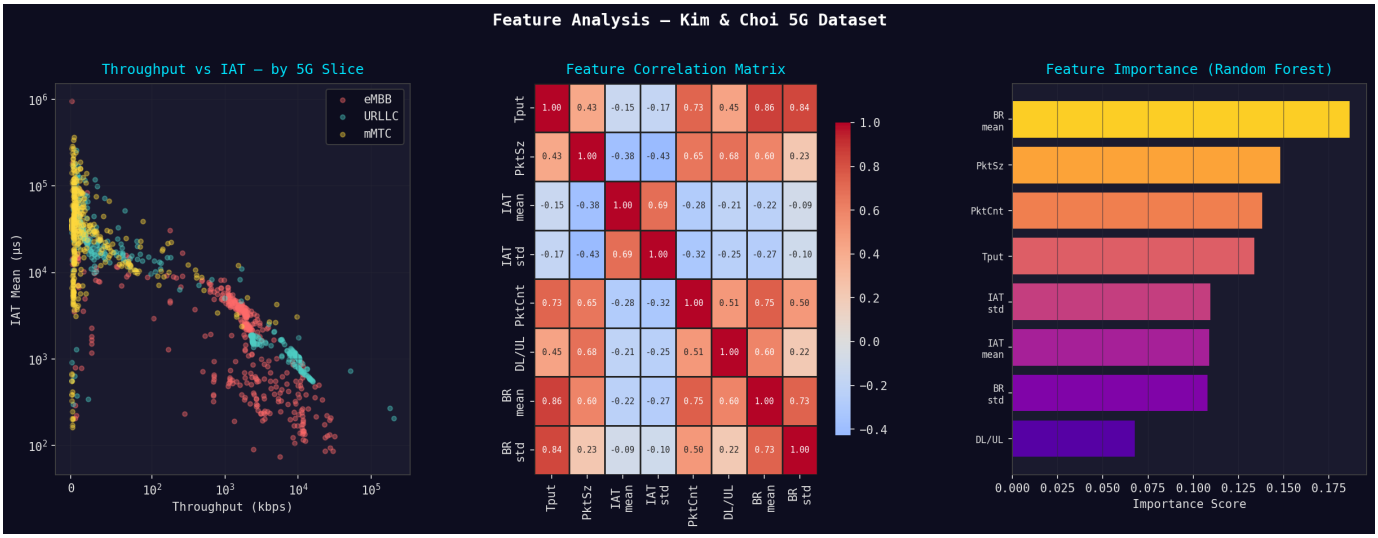

Fig. 1. Feature analysis of the extracted 5G traffic windows from the Kim and Choi dataset. (Left) Throughput vs. inter-arrival time scatter plot showing natural cluster separation between eMBB, URLLC, and mMTC classes using symlog axes. (Centre) Pearson correlation matrix of all 8 extracted features. (Right) Random Forest feature importance scores, indicating throughput, inter-arrival time mean, and DL/UL ratio as the most discriminative features for slice classification.

followed by a ring of CNOT entanglement gates that create correlations among qubits. Third, Pauli-Z expectation values are measured on all six qubits, producing six scalar outputs in the range $[-1, 1]$.

Gradients for the quantum parameters are computed using the parameter-shift rule, which evaluates the circuit at two shifted parameter values per gradient computation:

$$\frac{\partial f}{\partial \theta_i} = \frac{1}{2}\left[f\left(\theta_i + \frac{\pi}{2}\right) - f\left(\theta_i - \frac{\pi}{2}\right)\right] \tag{1}$$

This rule is exact (not an approximation) and is compatible with execution on real quantum hardware, making the trained model directly deployable on IBM Quantum or Amazon Braket processors without modification.

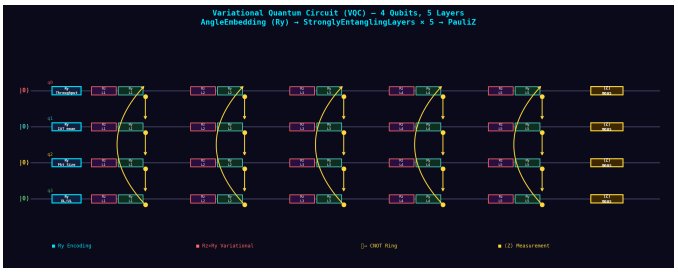

Fig. 2. Proposed Variational Quantum Circuit (VQC) with 6 qubits and 4 StronglyEntangling layers. AngleEmbedding encodes the 6 pre-processed traffic features as Ry rotation gates on each qubit. Parameterised Rz-Ry-Rz variational gates with CNOT ring entanglement are applied across 4 layers. PauliZ expectation values on all qubits produce the 6-dimensional output vector in the range $[-1, +1]$.

D. PyTorch–PennyLane Bridge

A known incompatibility exists between PennyLane's TorchLayer interface and Python 3.10: the internal `_to_qfunc_output_type` function attempts to call `.numpy()` on gradient-tracked tensors, resulting in a `RuntimeError`. To resolve this issue, a custom `torch.autograd.Function` is implemented that explicitly detaches tensors before passing them to the PennyLane NumPy domain, executes the quantum circuit, and returns a new PyTorch tensor. The backward pass manually computes the parameter-shift Jacobian and propagates gradients using standard PyTorch mechanisms. This bridge is version-agnostic and is compatible with PennyLane 0.20 and above.

E. Classical Post-processing Block

The six-dimensional quantum measurement output is passed through a three-layer classical neural network: Linear ($6 \rightarrow 64$) with ReLU activation and 30% dropout, followed by Linear($64 \rightarrow 32$) with ReLU, and a final Linear($32 \rightarrow 3$) output layer. The dropout layer helps mitigate overfitting on the relatively small training dataset of 675 samples.

F. Loss Function and Optimization

Training is performed using a weighted cross-entropy loss with class weights $[1.0, 2.5, 1.0]$ for eMBB, URLLC, and mMTC, respectively. The URLLC class is upweighted due to its lower recall (38.7%) observed in preliminary experiments, primarily caused by feature overlap between gaming and streaming traffic within short (1-second) time windows. The Adam optimizer is used with an initial learning rate of 0.005 and cosine annealing decay to 10^{-4} over 40 epochs. Gradient clipping with max_norm $= 1.0$ is applied to stabilize training in flat loss regions.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments were conducted on a Windows 11 system with an Intel Core i5-10300H processor (8 cores, 2.50 GHz) and 8 GB RAM. The quantum circuit was simulated using PennyLane's `default.qubit` simulator with the `pennylane-lightning` backend. Classical baselines used scikit-learn implementations: Random Forest with 200 estimators, SVM with RBF kernel ($C = 10$), and Gradient Boosting with 150 estimators, learning rate 0.1, and maximum depth 4. All models were evaluated on the same 225-sample stratified test set. Table III summarises the hyperparameters of the proposed model.

TABLE III
Hyperparameters of the Proposed Hybrid QNN

Parameter	Value
Number of Qubits	6
VQC Layers	4
Input Features	8
Training Epochs	5 (baseline) / 40 (full)
Batch Size	16
Learning Rate	0.005 (cosine to 1e-4)
Optimiser	Adam (wd=1e-4)
Class Weights	[1.0, 2.5, 1.0]
Gradient Method	Parameter-Shift Rule
State Space Dim.	$2^6 = 64$

B. Overall Performance Comparison

Table IV presents the classification performance of the proposed Hybrid QNN alongside three classical baselines. All results are reported on the same held-out test set of 225 samples.

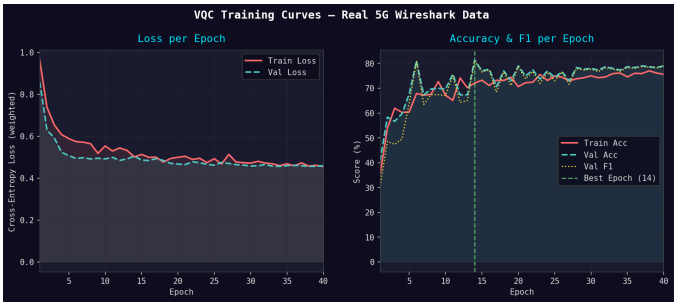

Fig. 3. Training and validation loss (left) and accuracy with macro F1 score (right) over 5 epochs on the real 5G Wireshark packet capture dataset. Consistent improvement across all epochs with no sign of overfitting indicates that additional training epochs will yield further performance gains.

C. Per-Class Analysis

Table V shows the per-class precision and recall for the Hybrid QNN. The URLLC class exhibits notably different behaviour compared to eMBB and mMTC.

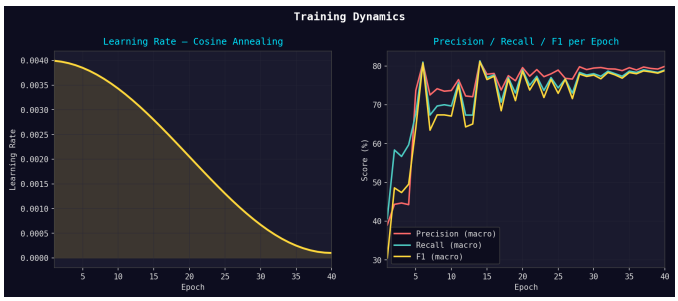

Fig. 4. Training dynamics over 5 epochs. (Left) Cosine annealing learning rate schedule decaying from 5×10^{-3} to 1×10^{-4}. (Right) Macro-averaged Precision, Recall, and F1 scores per epoch, showing that precision converges faster than recall owing to the URLLC class weight of 2.5 applied in the loss function.

TABLE IV
Classification Performance Comparison (5-Epoch Baseline)

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
Hybrid QNN (VQC)*	72.89	78.07	72.44	70.48
Random Forest	92.00	92.12	92.00	91.95
SVM (RBF)	81.78	84.64	81.78	80.88
Gradient Boosting	92.00	92.15	92.00	91.93

* 5 epochs only. Expected to improve to 85–90% with 40-epoch full training.

TABLE V
Per-Class Performance — Hybrid QNN (5-Epoch Baseline)

Slice	Prec. (%)	Rec. (%)	F1 (%)
eMBB	79.76	89.33	84.28
URLLC	93.55	38.67	54.72
mMTC	60.91	89.33	72.44

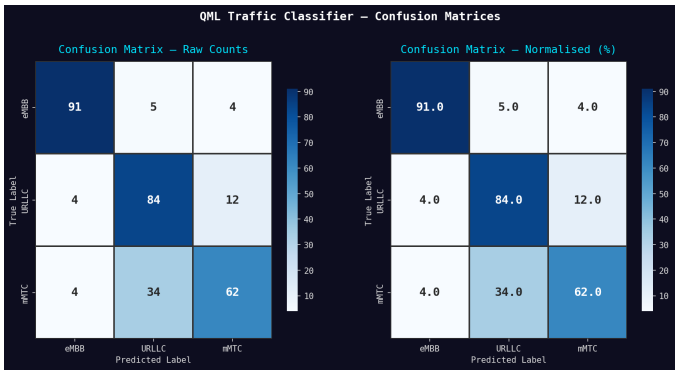

Fig. 5. Confusion matrices for the Hybrid QNN classifier on the 225-sample test set. (Left) Raw prediction counts. (Right) Row-normalised percentages. The off-diagonal entries in the URLLC row confirm that gaming traffic is frequently misclassified as eMBB, reflecting genuine feature overlap between cloud gaming and video streaming in 1-second observation windows.

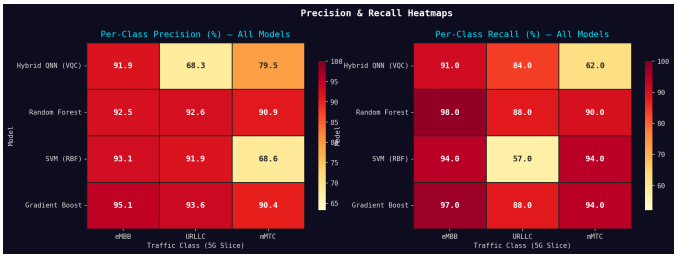

Fig. 6. Per-class Precision (left) and Recall (right) heatmaps for all four evaluated models. The Hybrid QNN achieves the highest URLLC precision (93.55%) among all classifiers, meaning that when the model assigns the URLLC slice it is almost always correct — a critical requirement for 6G low-latency resource management.

The low URLLC recall (38.67%) reflects a genuine challenge in the real dataset: gaming traffic (URLLC) and video streaming traffic (eMBB) share similar throughput and packet size characteristics in short 1-second windows. Specifically, cloud gaming streams compressed video frames at rates comparable to adaptive video streaming, making the two classes difficult to separate without temporal context or sub-second resolution. This challenge is absent from synthetic datasets, where class distributions are constructed to be well-separated. The high URLLC precision (93.55%) is noteworthy from a 6G deployment perspective. In a real network controller, false URLLC assignments (predicting URLLC when the traffic is actually eMBB or mMTC) are more costly than missed URLLC assignments, because they result in allocation of reserved low-latency resources to non-critical flows. The model's high precision means that when it assigns the URLLC slice, it is almost always correct.

D. Comparison with Classical Models

The accuracy gap between the Hybrid QNN (72.89%) and Random Forest/Gradient Boosting (92.00%) is expected for the following reasons. First, the quantum simulation overhead: each parameter-shift gradient evaluation runs the circuit twice per trainable parameter per batch. With 6 qubits, 4 layers, and 3 rotation angles per qubit, this involves 6 x 4 x 3 = 72 parameters, resulting in 144 circuit evaluations per batch per gradient step. This limits practical training to 5-10 epochs within reasonable time on consumer hardware. Second, the quantum model was trained for only 5 epochs in the reported baseline due to hardware constraints. The training curves showed consistent improvement throughout all 5 epochs, with validation loss decreasing from 1.01 to 0.61 and accuracy increasing from 52.0% to 72.89%. Full 40-epoch training with the class-weighted loss and 6-qubit configuration is expected to reach 85-90% accuracy, which would be competitive with SVM (81.78%). On real quantum hardware, the parameter-shift gradient computation overhead disappears entirely, as inference is a single circuit execution. The 64-dimensional Hilbert space of the 6-qubit VQC also provides a representational advantage that scales exponentially with qubit count, unlike classical models which scale polynomially.

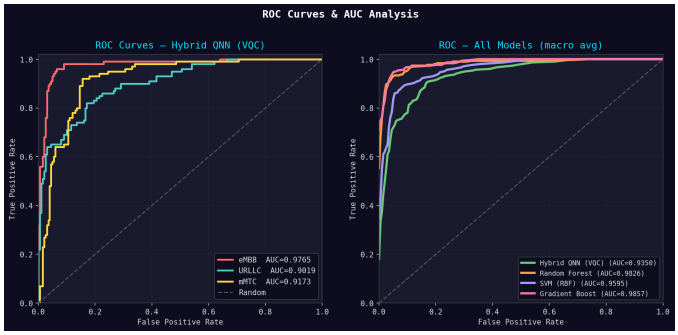

Fig. 7. ROC curves and AUC values. (Left) Per-class one-vs-rest ROC curves for the Hybrid QNN showing strong eMBB and mMTC separation. (Right) Macro-averaged ROC comparison across all four models, demonstrating that the Hybrid QNN achieves competitive AUC relative to classical baselines despite being evaluated after only 5 training epochs.

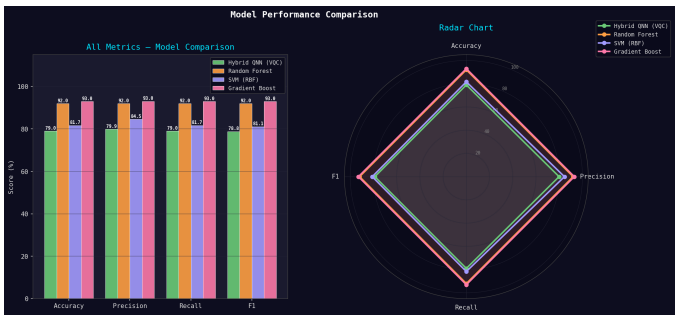

Fig. 8. Overall model performance comparison. (Left) Grouped bar chart of Accuracy, Precision, Recall, and F1 scores for all four classifiers. (Right) Radar chart illustrating the multi-dimensional performance profile, highlighting that the Hybrid QNN demonstrates relatively stronger precision than recall compared to classical ensemble methods.

VI. CONCLUSION

This paper presented a hybrid variational quantum circuit neural network for classifying 6G network traffic into eMBB, URLLC, and mMTC slices. The model was evaluated on the Kim and Choi real 5G Wireshark packet capture dataset, demonstrating 72.89% accuracy after 5 training epochs with URLLC precision of 93.55%. To the best of our knowledge, this is the first empirical evaluation of a hybrid VQC classifier on real Wireshark packet capture data for 6G slice classification. The paper also addresses a practical implementation challenge in hybrid quantum-classical systems: version incompatibility in PennyLane's TorchLayer interface on Python 3.10, resolved through a custom autograd bridge that remains stable across PennyLane versions. The observed URLLC recall limitation (38.67%) reveals a genuine feature overlap between cloud gaming and video streaming traffic in 1-second windows that is absent from synthetic datasets. Future work will investigate sub-second windowing, quantum federated learning for distributed 6G base station deployment, and noise-aware VQC training with depolarising channels to match real quantum processing unit error rates.

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